MILITARY SPECIFICATION

PROJECTILE, PRACTICE, 155MM, M804E1
CAST METAL PARTS ASSEMBLY

This Specification is approved for use by the U.S. Army Armament, Munitions and Chemical Command, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the Metal Parts Assembly for the projectile, Practice, 155MM, M804E1, Cast Body (See 6.1).

2. APPLICABLE DOCUMENTS

2.1 Government documents

2.1.1 Specifications, standards and handbooks. The following specifications, standards and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (See 6.2).

SPECIFICATIONS

FEDERAL

TT-C-490 - Cleaning Method and Pre-treatment of Ferrous Surfaces for Organic coatings

MILITARY

MIL-A-2550 - Ammunition, General Specification for
MIL-A-48078 - Ammunition, Standard Quality Assurance Provisions, General Specification for

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, U.S. Army ARDEC, ATTN: SMCAR-BAC-S, Picatinny Arsenal, New Jersey 07806-5000 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A
FSC 1320

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STANDARDS

MILITARY

MIL-STD-410 - Non-destructive Testing Personnel Qualification and Certification
MIL-STD-453 - Inspection, Radiographic

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from: Standardization Documents Order Desk, Bldg. 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

DRAWINGS (see 6.9)

U.S. Army Armament Research, Development and Engineering Center (ARDEC)

7548993 - Blank, Rotating Band
12953254 - Packaging and Marking for Pallet, Wood, Ammunition, for Projectile, 155MM
12913926 - Projectile, 155MM Practice, M804E1 Loading, Assembly and Marking diagram
12913927 - Projectile, 155MM Practice, M804E1 Metal Parts Assembly
12913928 - Body Assembly
12913929 - Body

PUBLICATIONS

FRANKFORD ARSENAL

Operating Instructions for Use of Detroit Testing Machine for Band Seat Inspection.

(Copies of drawings and other Government documents required by contractors in connection with specific acquisition functions should be obtained from the Contracting activity or as directed by the Government Officer.)

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2.2 Non-Government publications. The following document(s) form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents cited in the solicitation (See 6.2).

AMERICAN NATIONAL STANDARDS INSTITUTE (ANSI)

ANSI B46.1 - Surface Texture

(Application for copies should be addressed to the American National Standards Institute (ANSI), 1430 Broadway, N.Y., N.Y. 10018).

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM A-247	- Evaluating the Microstructure of Graphite in Iron Castings
ASTM B-117	- Method of Salt Spray (Fog) Testing
ASTM E-3	- Metallographic Specimens
ASTM E-8	- Tension Testing of Metallic Materials
ASTM E-10	- Brinell Hardness of Metallic Materials
ASTM E-103	- Rapid Indentation Hardness Testing of Metallic Materials
ASTM E-446	- Reference Radiographs Vol. II for Cast Steel

(Application for copies should be addressed to the American Society for Testing and Materials,1916 Race Street, Philadelphia, PA. 19103).

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Material. Materials, parts and assemblies shall comply with the requirements specified on the applicable drawings and specifications.

3.2 Projectile assembly. The projectile assembly and components shall comply with all requirements specified on Drawing 12913928 and with all requirements specified in applicable specifications.

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3.3 Material quality. The contractor shall establish procedures to provide material in accordance with the applicable ASTM specifications (see 2.2). This material shall be free of pipe, porosity and heavy segregation.

3.4 Process Control Document (PCD). A process control document covering the melting, casting and mold cooling operations shall be submitted to the technical agency through the contracting officer for review and approval prior to initiation of production. This document shall include descriptions of procedures, equipment, process parameters, monitoring practices, calibration procedures and frequencies of testing for all listed control parameters. The process control document shall be submitted in accordance with DI-P-1604. Any revisions to the PCD shall be submitted to the technical agency through the contracting officer for review and approval.

As a minimum, the following process parameters shall be addressed in the PCD:

a. Melt furnace and ladle chemistry.
b. Inoculation practice.
c. Casting pour temperature.
d. Time from inoculation to last metal poured.
e. Time from pour to shakeout.
f. Nodularity control.
g. Mold & core fabrication and materials.
h. Pattern design and chill practices, if any.
i. Core stabilization practice, if used.

The documentation/record keeping procedures to be used for recording all parameter measurements shall also be detailed in the process control document.

3.5 Mechanical properties.

3.5.1 Body. Mechanical properties of the machined body as required by applicable drawing shall apply.

3.5.2 Rotating band. After application and subsequent removal the rotating band shall have the following mechanical properties:

a. Tensile Strength - 206.8 Megapascals (MPa) (30,000 psi) minimum (min)
b. Elongation - 25 percent in 20 millimeters (mm) or .75 inch (in) of gage length min.

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3.6 Rotating band.

3.6.1 Band seat. Prior to banding, the band seat shall be clean and free from oil, grease, dirt, rust or other foreign matter. Shot or sandblasting shall not be permitted for cleaning the band seat after knurling.

3.6.2 Band blank. (Gilding metal). The rotating band blank shall be in the form of a solid ring, annealed if necessary to comply with the applicable specifications; except that when hot banding at not less than 427 degrees Celsius (C), no prior annealing is necessary.

3.6.3 Band seating sidewall contact. The rotating band shall be applied by carefully pressing the band blank into the band seat so as to make contact with the sidewalls for a minimum of eight-tenths of the band seat depth. The band shall be applied in such a manner so as not to distort the body wall. In addition, there shall be no surface gap between the rotating band and the sidewalls of the band seat, after the rotating band has been machined to final dimension.

3.6.4 Band seating tightness. The diametrical clearance between the rotating band and the band seat shall not exceed 0.15mm (.006 in), when the destructive method of testing is used or 0.23mm (.009 in) when non-destructive method of testing is used.

3.7 Metal defects.

3.7.1 Body. The body, including the cavity, shall be free from cracks, splits, bursts, cold shuts, pipes, porosity, inclusions, folds, seams and other metal defects.

3.7.2 Visual radiographic standards. Visual standards for evaluation of metal defects will be established by the contractor with guidance from the Government Agency for each contract after the first months production. However, establishment of visual standards will not waive the requirements for compliance with 3.7.1.

3.7.3 Non-destructive testing. Non-destructive testing shall be performed in the following order:

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a. Hydrostatic test. Projectile bodies shall withstand an internal hydrostatic pressure of 10,000 + 500 pounds per square inch (psi) for 15 seconds minimum. There shall be no evidence of leakage, cracks, ruptures or permanent distortion when tested in accordance with 4.5.6.

b. Radiographic inspection. Each body shall be radiographed in accordance with 4.5.5 and meet the requirements of ASTM E-446 with the following exceptions:

1. There shall be no indications in the following categories; A, B, D, E, F, G.

2. Level I or better for the following categories; CA, CB, and CD.

3. Level I or better for category CC - shrink in the area illustrated in figure 1. No worse than level III, of category CC, in the remaining areas.

Only film as listed in 6.4 shall be used for radiography of this projectile.

3.8 Tool control dimensions. Dimensions marked tool control shall be gaged at the beginning of production and whenever tooling is replaced. Where destruction of components is necessary to inspect these dimensions, measurements of the tool may be substituted provided the contractor has established correlation between the tool dimensions and the component dimensions prior to start of production. The contractor shall be responsible for maintaining adequate surveillance to assure that drawing requirements are being met throughout the life of the tool.

3.9 Microstructure. The microstructure of the body shall be examined to determine form, distribution and size of graphite particles.

3.10 Workmanship. The requirements for workmanship are as shown on the applicable drawings and specifications and the following:

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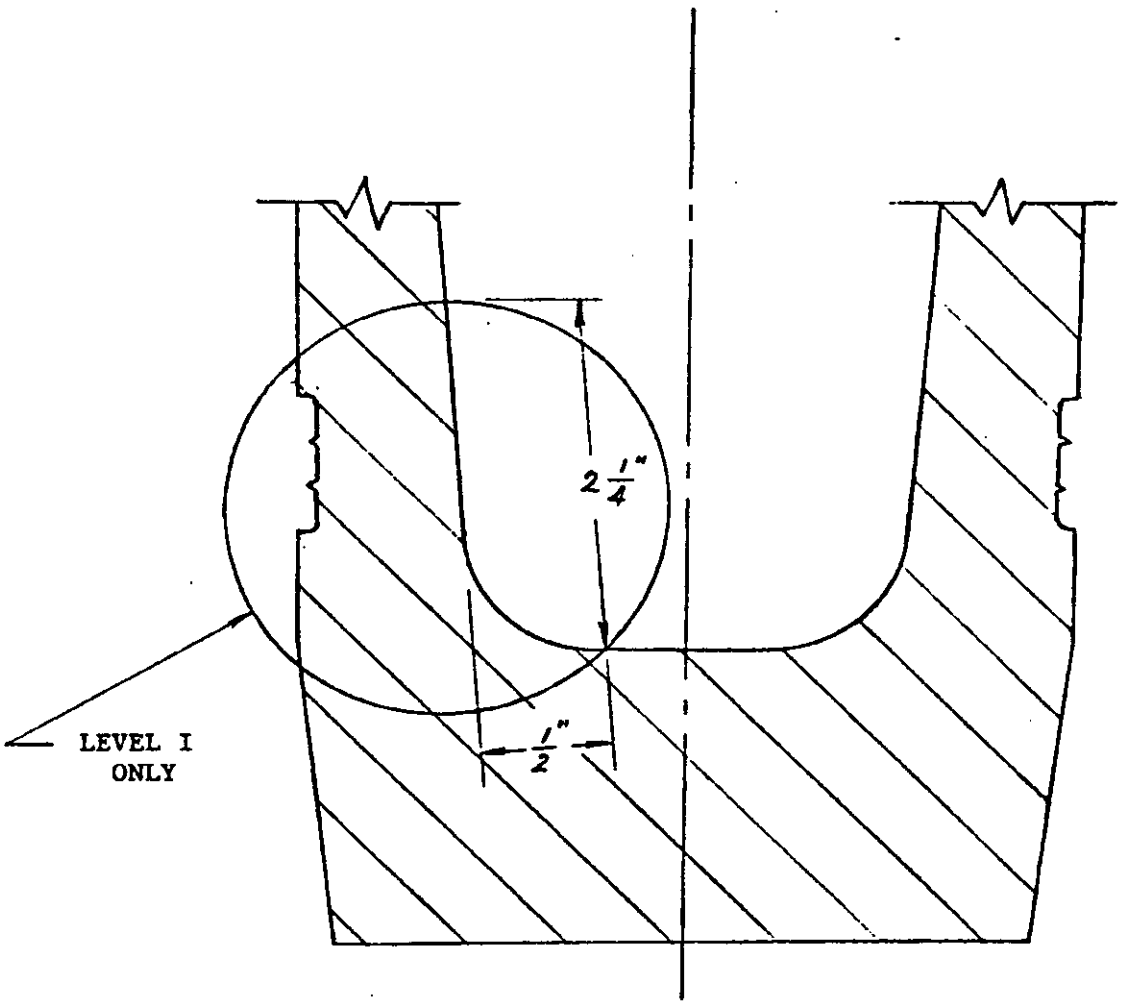


FIGURE 1. HIGH STRESS AREA

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3.10.1 Surface preparation. Preeparation of metal surfaces shall be as specified on the applicable drawings and specifications. Cleaning and preparation are required by component details, the finish specified shall apply to indicated surfaces regardless of contour configuration.

3.10.2 Painting. Painting shall comply with the requirements of the applicable drawings and specifications. All paint shall be dry to the touch before packing for shipment. Drying time prior to testing shall be in accordance with the applicable specification.

3.10.3 Surface finish. The requirements for surface finish are as detailed on the applicable drawing. The roughness comparison specimens prescribed in ANSI-B46.1 shall be used as a basis for surface roughness determinations.

3.10.4 Threads. Threads shall be full and undamaged for the entire minimum length or depth as spcified on the applicable drawing.

3.10.5 Burr. No part shall have a burr which might interfere with the assembly or function of the projectile or which might be injurious to personnel handling the item.

3.10.6 Foreign matter. No part or assembly shall contain dirt, grease, rust, corrosion or other foreign mater.

3.10.7 Protection of rotating band. The machined rotating band shall be protected from being nicked, burred or otherwise damaged in subsequent handling prior to shipment.

3.11 Proving ground. There shall be no break-up or separation of metal parts in the gun bore or in flight, when tested.

3.12 First article inspection. This specification makes provisions for first article inspection. Requirements for the submission of first article sample by the contractor shall be specified in the contract.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items must meet all requirements of Section 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

4.2 Classification of inspections. The following types of inspection shall be conducted on this item:

a. First article inspection

b. Quality conformance inspection

4.3 First article inspection.

4.3.1 Inspection and tests (non ballistic).

4.3.1.1 Submission. The contractor shall submit a first article sample as designated by the Contracting Officer for evaluation in accordance with the provisions of 4.3.2. The first article sample shall consist of the following items in sample quantities as indicated.

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Part Description	Drawing	Quantity
Blank, Rotating Band	7548993	5
Body (unpainted)	12913929	5
Body Assembly	12913928	5
Projectile Metal Parts	12913927	45

All assemblies and components shall have been produced by the contractor or furnished by a supplier and shall have been manufactured using the same production processes, procedures and equipment which will be used in fulfilling the contract. All parts and materials, including packaging and packing, shall be obtained from the same source of supply as will be used in regular production. Prior to submission, the contractor shall inspect the sample to the degree necessary to assure that it conforms to the requirements of the contract and submit a record of this inspection with the sample including statements of findings for materials, processes and tests. A sample containing known defects will not be submitted unless specifically authorized, by the Contracting Officer. A first article sample, or portion thereof, as directed by the Contracting Officer, shall also be submitted whenever there is a lapse in production for a period in excess of 90 days or whenever a change occurs in manufacturing process, material used, drawing or specification such as to significantly affect product uniformity determined by the Government.

4.3.2 Inspections to be performed. See MIL-A-48078 and Table I specified herein.

4.3.2.1 Ballistic tests.

4.3.2.1.1 Submission. Ten complete assemblies packed in the form that will be used for shipping of the regular production quantity shall be submitted to a Government installation designated by the Contracting Officer for loading and ballistic testing. Unless otherwise directed by the Contracting Officer, the assemblies shall not be submitted prior to acceptance of the sample submitted to the Government for examination (see 4.3.1) and shall consist of ten assemblies.

4.3.2.1.2 Tests to be performed and acceptance criteria. Proving ground tests will be performed on ten assemblies in accordance with 4.4.3.1, using the procedures of 4.5.8 to determine compliance with 3.10.

4.3.3 Rejection. See MIL-A-48078.

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TABLE I - First article inspection

CLASSIFICATION OF CHARACTERISTICS

MIL-P-70695 (AR)

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
	Projectile Assembly and Components	1 OF 2	See Below
			NEXT HIGHER ASSEMBLY

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
	Blank, Rotating Band			
	Dwg. 7548993			
	Mechanical Properties	5; 100%	3.5	4.5.2
	Body (unpainted)			
	Dwg. 12913929			
	Examination for Defects	5; 100%	3.1	4.4.2.1
	Mechanical Properties	5; 100%	3.5	4.5.2
	Microstructure	5; 100%	3.9	4.5.7
	Hydrostatic	5; 100%	3.7.3a	4.5.6
	Radiography	5; 100%	3.7.3b	4.5.5
	Body Assembly			
	Dwg. 12913928			
	Examination for Defects	5; 100%	3.1	4.4.2.2
	Band Seating	5; 100%	3.6	4.5.3.1
	Mechanical Properties Rotating Band	5; 100%	3.5.2	4.5.3.2.1

NOTES:

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TABLE I - First article inspection

CLASSIFICATION OF CHARACTERISTICS

MIL-P-70695 (AR)

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
	Projectile Assembly and Components	2 OF 2	See Below
			NEXT HIGHER ASSEMBLY

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
	Projectile Metal Parts Assembly			
	Dwg. 12913927			
	Examination for defects	45; 100%	3.2	4.4.2.3
	Salt Spray Testing	45; 100%	3.1	4.5.4.2
	Protective Coating Thickness	45; 100%	3.10.2	4.5.4.1
	Functional Test (Proving Ground)			
	Dwg. 12913926			
	Ballistic Testing	10; 100%	3.11	4.5.8

NOTES:

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4.4 Quality conformance inspection.

4.4.1 Inspection lot formation. Inspection lots shall comply with the formation provisions of MIL-A-48078 and the following:

4.4.1.1 Casting lot size. The casting lot is defined as the following:

a. Until five consecutive casting lots pass mechanical properties, a lot shall be a maximum of 100 projectile bodies. If initial casting lots are smaller than 100 bodies, those lots representing 500 consecutively cast bodies shall meet mechanical properties before use of casting lots larger than 100 bodies per lot can begin.

b. After 500 consecutive projectile bodies pass mechanical property requirements, the casting lot size shall be the casting production from one shift or 1000 projectiles, whichever is smaller. Failure of any lot to meet mechanical property requirements shall be cause for rejection of the lot and require a return to the practice described in paragraph a.

4.4.2 Examinations and tests.

a. Classification of characteristics. Quality conformance examinations and tests are specified in the following Classification of Characteristics paragraphs. The contractor's quality program or detailed inspection system shall provide assurance of compliance of all characteristics with the applicable drawing and specification requirements utilizing as a minimum the conformance criteria specified. When cited herein, attributes sampling inspection shall be conducted in accordance with Table II below, using the inspection levels stated in the Classification of Characteristics paragraphs.

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TABLE II . Attributes sampling inspection

Lot Size	Inspection Levels					
	I	II	III	IV	V	VI
2 to 8	*	*	*	*	5	3
9 to 15	*	*	*	13	5	3
16 to 25	*	*	*	13	5	3
26 to 50	*	*	32	13	5	3
51 to 90	*	*	32	13	13	5
91 to 150	*	125	32	13	13	5
151 to 280	*	125	32	32	20	8
281 to 500	*	125	32	32	20	8
501 to 1200	*	125	80	50	20	13
1201 to 3200	1250	125	80	50	32	13
3201 to 10000	1250	125	125	50	32	13
10001 to 35000	1250	315	125	80	50	13
35001 to 150000	1250	315	125	80	50	13
150001 to 500000	1250	500	200	125	50	13
500001 and over	1250	500	200	125	50	13

Numbers under inspection levels indicate sample size; asterisks ( ) indicate one hundred percent inspection. If sample size exceeds lot size, perform one hundred percent inspection. Accept on zero and reject on one or more for all inspection levels.

b. Alternative quality conformance provisions. Alternative quality conformance procedures, methods, or equipment, such as statistical process control, tool control, other types of sampling procedures, etc., may be used by the contractor when they provide, as a minimum, the level of quality assurance required by the provisions specified herein. Prior to applying such alternative procedures, methods, or equipment, the contractor shall describe them in a written proposal submitted to the Government for evaluation (see 6.11). When required, the contractor shall demonstrate that the effectiveness of each proposed alternative is equal to or better than the specified quality assurance provision(s) herein. In cases of dispute as to whether the contractor's proposed alternative(s) provides equivalent assurance, the provisions of this specification shall apply. All approved alternative provisions shall be specifically incorporated into the contractor's quality program or detailed inspection system, as applicable.

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QUALITY CONFORMANCE INSPECTION
CLASSIFICATION OF CHARACTERISTICS

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.1	Body	1 OF 2	12913929/12913928
			NEXT HIGHER ASSEMBLY: 12913927

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Critical	None defined			
Major				
101	Metal defective	100%, 0-1	3.7	Visual
102	Run out of rear face	Level III	3.2	Gage
103	Depth of scoring	Level III	3.2	Visual
104	Scoring or knurling missing from rotating band seat	Level III	3.2	Visual
105	Run out of pitch diameter of thread	Level III	3.2	Gage
106	Run out of ogive	Level III	3.2	Gage
107	Run out of bourrelet	Level III	3.2	Gage
108	Run out of boattail	Level III	3.2	Gage
109	Variation in wall thickness	Level III	3.2	Gage
110	Pitch diameter of thread, max.	Level III	3.2	Gage
111	Minor diameter of thread, max.	Level III	3.2	Gage
112	Length of effective thread, min.	Level III	3.2	Gage
113	Mechanical properties of body	4.4.3.1	3.5	4.5.2
114	Length of effective thread, max.	Level III	3.2	Gage
115	Radiographic inspection	4.5.5	3.7.3b	4.5.5
116	Hydrostatic test	100%	3.7.3a	4.5.6
117	Microstructure data missing	4.4.3.5	3.9	4.5.7

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF CHARACTERISTICS

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.1	Body (con't)	2 OF 2	12913429/12913928
			NEXT HIGHER ASSEMBLY
			12913927

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Minor				
201	Radii or chamfers missing or incorrect	Level V	3.2	Visual
202	Surface finish improper	Level V	3.10.3	Visual
203	Evidence of shot or sandblasting on rotating band seat	Level V	3.2	Visual
204	Burr	Level V	3.10.5	Visual
205	Diameter of thread chamfer	Level V	3.2	Gage
206	Diameter at rear of boattail	Level V	3.2	Gage
207	Length from rear face to rear of bourrelet	Level V	3.2	Gage
208	Diameter of rotating band seat	Level V	3.2	Gage
209	Location of rotating band seat	Level V	3.2	Gage
210	Width or rotating band seat	Level V	3.2	Gage
211	Overall length	Level V	3.2	Gage
212	Length from front of boattail to rear of rotating band seat	Level V	3.2	Gage
213	Thickness through base	Level V	3.2	Gage
214	Diameter of bourrelet, min.	Level V	3.2	Gage
215	Diameter of body	Level V	3.2	Gage
216	Length from rear face to base diameter	Level V	3.2	Gage
217	Thickness of body wall, min.	Level V	3.2	Gage
218	Inside diameter of nose, below thread	Level V	3.2	Gage
219	Evidence of poor workmanship	Level V	3.10	Visual

NOTES:

AMSC N/A Form 1570b, 1 Jul 89 Replaces 1570, 1 Feb 85, which may not be used.

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF CHARACTERISTICS

MIL-P-70695 (AR)

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.2	Body Assembly	1 OF 2	12913928
			NEXT HIGHER ASSEMBLY 12913927

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Critical	None defined			
Major				
101	Runout of rotating band	Level III	3.2	Gage
102	Diameter of rotating band	Level III	3.2	Gage
103	Location of rotating band	Level III	3.2	Gage
104	Excessive gap between rotating band and side wall	4.4.3.2	3.6.3	4.5.3
105	Excessive gap between rotating band and seat	4.4.3.3	3.6.4	4.5.3
106	Rotating band mechanical properties	4.4.3.2	3.5	4.5.3.2
107	Rotating band seat not clean, prior to application	Level III	3.10.6	Visual
Minor				
201	Protective coatings or paint missing incomplete, or improper	Level V	3.2	Visual
202	Marking missing	Level V	3.2	Visual
203	Length, max	Level V	3.2	Gage
204	Diameter of rotating band groove	Level V	3.2	Gage
205	Length from rotating band groove to start of angle	Level V	3.2	Gage
206	Length of rotating band flat	Level V	3.2	Gage
207	Angle at front of rotating band	Level V	3.2	Gage
208	Angle at rear of rotating band relief groove	Level V	3.2	Gage

NOTES:

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QUALITY CONFORMANCE INSPECTION
CLASSIFICATION OF CHARACTERISTICS

MIL-P-70695 (AR)

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.2	Body Assembly (con't)	2 OF 2	12913928
			NEXT HIGHER ASSEMBLY 12913927

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
209	Width of rotating band	Level V	3.2	Gage
210	Width of rotating band groove	Level V	3.2	Gage
211	Radii or chamfers missing or incorrect	Level V	3.2	Visual
212	Width of rotating band relief groove	Level V	3.2	Gage
213	Assembly weight incorrect	Level V	3.2	Scale
214	Stencil incorrect	Level V	3.2	Visual
215	Evidence of poor workmanship	Level V	3.10	Visual

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF CHARACTERISTICS

MIL-P-70695 (AR)

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.3	Projectile, 155mm, Practice, M804 Metal Parts Assembly	1 OF 1	12193927 NEXT HIGHER ASSEMBLY

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Critical	None defined			
Major				
101	Rotating band damaged, prior to assembly of grommet	Level III	3.10.7	Visual
102	Protective coating thickness	4.4.3.4	3.10.2	4.5.4
103	Diameter of bourrelet, max.	100%	3.2	Gage
Minor				
201	Corrosion preventive compound missing or improperly applied	Level V	3.2	Visual
202	Evidence of poor workmanship	Level V	3.10	Visual

NOTES:

AMSMC Form 1570b, 1 Jul 89 Replaces 1570, 1 Feb 85, which may not be used.

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF CHARACTERISTICS

MIL-P-70695 (AR)

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.4	Packaging and Marking	1 OF 1	9331807/9334433
			NEXT HIGHER ASSEMBLY

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Critical	None defined			
Major				
101	Pallet cover/base damaged or improperly assembled	Level III	3.2	Visual
102	Strapping or wire missing or loose	Level III	3.2	Visual
Minor				
201	Number of projectile assemblies incorrect	Level V	3.2	Visual
202	Marking misleading, unidentifiable or incorrect	Level V	3.2	Visual
203	Strapping or wire improperly engaged	Level V	3.2	Visual
204	Evidence of poor workmanship	Level V	3.10	Visual

NOTES:

DLIC Form 1570b, 1 Jul 89 Replaces 1570, 1 Feb 85, which may not be used.
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4.4.3 Testing.

4.4.3.1 Mechanical properties of body.

4.4.3.1.1 Hardness. In order to determine uniformity of heat treatment, each body in the casting lot shall be subjected to a hardness test approved by the procuring activity.

4.4.3.1.2 Tension test. Two projectile bodies, one of the hardest and one of the softest as determined by the hardness test, shall be subjected to this test. Failure of any tension test requirements of the applicable drawing shall be cause for rejection of the lot.

4.4.3.1.3 Dynamic tear test. Beginning with the first melt produced and continuing until three consecutive melts have complied with the applicable minimum requirement specified, one projectile from each melt shall be subjected to the dynamic tear test. The projectile selected shall be of the highest hardness reading from 4.4.3.1.1 and may be the same as used in 4.4.3.1.2. Two specimens shall be tested, with acceptance based on the average fracture toughness. In addition, no individual total foot-lbs of energy shall be less than 4.5 ft-lbs for dynamic tear sample. If one test is invalid, acceptance may be based on the total energy of the valid sample, or another sample from the same projectile may be tested. If either the average or individual total energy is below the minimum allowable, the melt represented shall be rejected. After three consecutive melts have been accepted, the test shall be discontinued. The test shall be reinstituted until three consecutive melts have complied with the applicable requirements, if any of the following occur:

a. Change in melt mix.
b. Change in material vendor.
c. Change in solidification (shake out time).

4.4.3.2 Rotating band mechanical properties and band seating sidewall contact. One body assembly shall be selected from each 8-hour production or fraction thereof from each banding machine in operation to determine compliance with applicable requirements. Failure to comply shall be cause for rejection of the assemblies represented by the sample.

4.4.3.3 Band seating tightness.

4.4.3.3.1 Non-destructive method. Each body assembly shall be subjected to this test. Failure to comply with applicable requirements shall be cause for rejection of that body assembly.

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4.4.3.3.2 Destructive method. When the equipment for the non-destructive method of testing is not available, one body assembly shall be selected from each 8-hour production or fraction thereof per banding machine in operation. Failure of any band to comply with applicable requirements shall be cause for rejection of the production represented by the sample.

4.4.3.4 Protective coating - thickness. Two samples shall be selected at random to represent each four hour period, or less, of continuous paint production. The sample projectiles shall be inspected for paint coating thickness at a minimum of four separate locations on each projectile. Any projectile failing to meet the applicable requirement shall be classed defective and every projectile painted since the last acceptable sample shall be rejected.

4.4.3.5 Protective coating - salt spray. The sample size and acceptance criteria shall be as specified in the applicable specification. If failure occurs, all items processed since last acceptance shall be rejected and corrective action taken.

4.4.3.5.1 Salt spray. After qualification in accordance with TT-C-490, two (2) salt spray samples shall be taken each week, one during the first 30 minutes of production and a second at any other time. Failure to comply with the requirements of the applicable drawing shall be cause for taking the corrective action prescribed in TT-C-490.

4.4.3.6 Microstructure. An examination per ASTM A-247 shall be made of the microstructure of each casting lot to determine form, distribution and size of graphite particles.

4.4.4 Inspection equipment. The contractor shall submit for approval inspection equipment designs in accordance with the terms of the contract. See Section 6 of MIL-A-48078 and 6.3 herein.

4.5 Methods of inspection.

4.5.1 Metal defects. The projectile shall be inspected visually using peripheral lights to assure conformance with the applicable requirements. Visual standards, in accordance with applicable requirements shall be set up for use during this inspection. Any projectile failing to meet the applicable requirements shall be classed defective and removed from the lot.

4.5.2 Mechanical properties of body.

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4.5.2.1 Hardness. Prior to final machining, a Brinell hardness test shall be performed on the outside surfaces at the band seat location or at any other area on the body deemed necessary for process control. The equipment and procedures for performing this test are prescribed by ASTM Method E-103 and the following:

a. The surface of the body in the area to be hardness tested shall be prepared by machining or grinding to a sufficient depth, .015 inches minimum, to remove scale and get below the decarburization zone. Manual grinding shall not be used.

b. Hardness tester periodic verification shall be performed using Brinell test standards manufactured in accordance with ASTM Method E-10 in lieu of production samples.

c. All hardness test measurements shall be automatically displayed.

d. The hardness tester shall use a standard 10MM Brinell ball. The body shall be securely supported so that no rocking or shifting of the body will occur during the test. The tester shall apply a preload followed by application of the full load. The preload shall not exceed 10% of the major load of 3000Kg. The amount of penetration of the indentor (10MM ball) during that time period when the major load is applied including the dwell time at the end of the load application shall be automatically measured. The tester shall be provided with a means to compensate for any flexing of the test item that may occur during application of the major load. The hardness number will be displayed at the conclusion of the test.

4.5.2.2 Tension test. From the sample projectiles selected, two tension specimens conforming to the largest obtainable standard round type specimens shall be tested in accordance with the procedure prescribed by ASTM E-8. The specimens shall be taken longitudinally, from opposite sides of the body in such a manner that the center of the gage length of the machine test bar will fall in line coincident with the center of the rotating band seat. The number of specimens and their locations are for the convenience of the contractor and Government and shall not restrict the contractor or Government from testing additional specimens at other locations to determine that the mechanical properties of the projectile meet the drawing requirements. The criteria for performing the tests shall be as prescribed by ASTM E-8.

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4.5.2.2.1 Nose section. One longitudinal specimen shall be taken from the ogive, not more than 76mm (3 in) from front face. Failure to meet the yield strength or elongation requirement shall be cause for rejection of the inspection lot.

4.5.2.2.2 Dynamic tear test. The two specimens shall be taken from locations D1 and D2 on figure 2. The test shall be conducted at a specimen temperature of -60 degrees Fahrenheit (F). The test shall be conducted in accordance with ASTM E604-83 (reapproved 1988). Location D3 and D4 of figure 2 shall be used for replacement specimens for locations D1 and D2 respectively.

4.5.3 Rotating band.

4.5.3.1 Band seating tightness.

4.5.3.1.1 Nondestructive method. Each sample shall be tested utilizing the equipment and procedures prescribed in Frankford Arsenal Detroit Testing Machine Manual (see 2.1), for test of 155MM, Proj., M107.

4.5.3.1.2 Destructive method. The body assembly selected should not be processed through the band machining line, although each band should have a light surface cut to true up the surface prior to measuring. The diameter of each band shall be measured at 3 locations, 60 degrees apart, on the points along the cylindrical portion of their length, near the front and rear edges. This comprised 6 diametrical measurements per band. These points of measurements shall be marked on the bands and on the projectile body adjacent thereto in such a manner as to permit proper body and band orientation after removal. The bands shall be carefully removed so as to keep deformation to a minimum. After removal, the thickness of the band shall be measured at each of the marked points (a total of 12 measurements) and the diameter of the band seat corresponding to these points shall also be measured. Add to each of the diameters of the band seat the sum of the corresponding 2 thicknesses of the band; the results obtained shall then be subtracted from the previously determined diameters over the band.

4.5.3.2 Mechanical properties and sidewall contact. The samples shall be sectioned by cutting through two planes normal to the longitudinal axis one centimeter forward and rear of the rotating band. The cylindrical portion containing the rotating band its longitudinal axis. One of the two halves shall be used for inspection for mechanical properties and the other section for inspection of sidewall contact.

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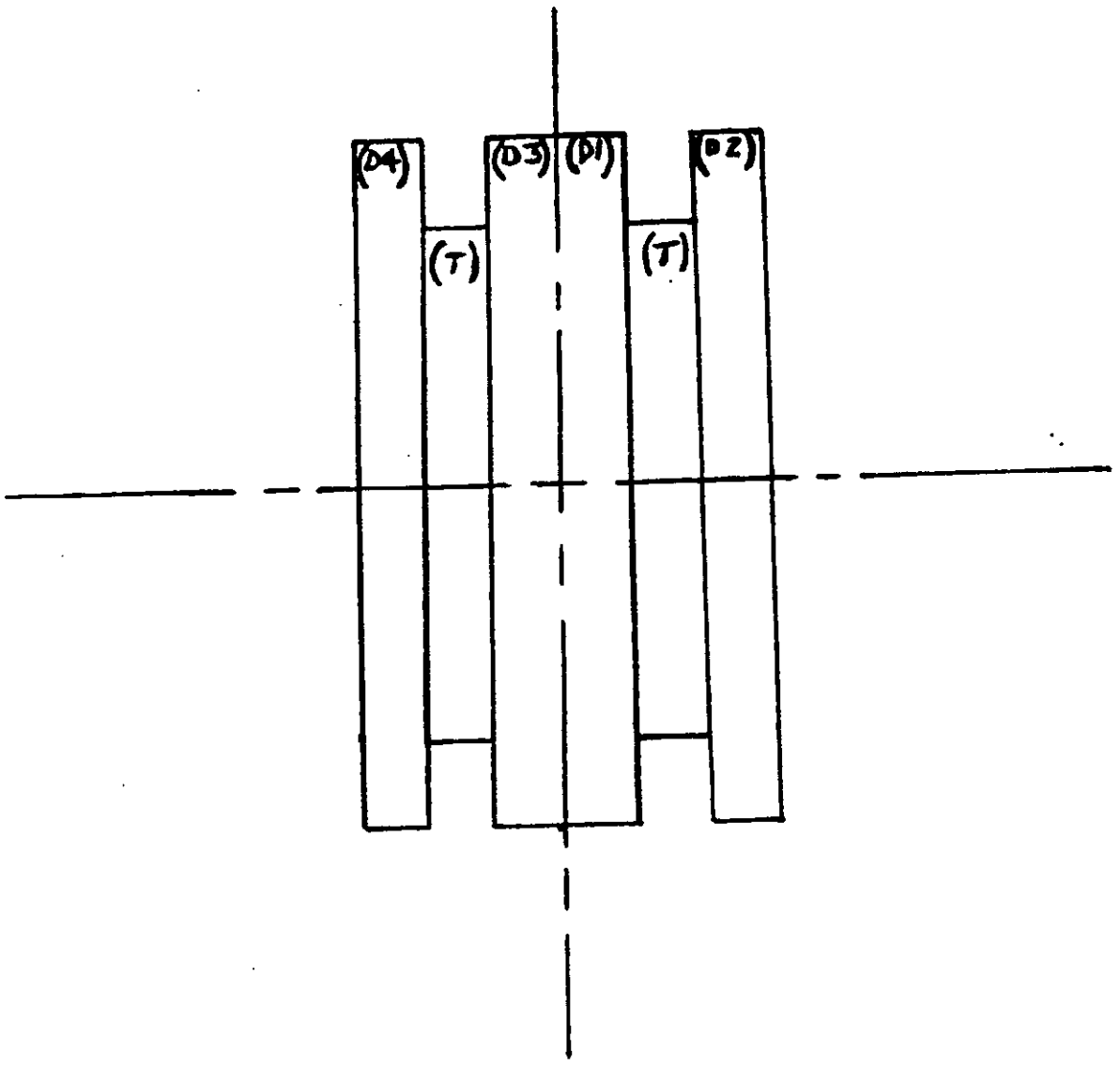


Figure 2. Dynamic tear test bars.

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4.5.3.2.1 Mechanical properties. The section provided shall be flattened without hammering so as to hold to a minimum any additional cold working. The specimen blank shall be machined so that all serration marks from the band seat are removed. a test specimen having two centimeter gage length to accomodate the tensile machine, shall be removed from the blank band. The specimen shall be tested in accordance with the procedure prescribed by ASTM Method E-8.

4.5.3.2.2 Sidewall contact. The specimen provided shall be sectioned into two parts exposing four surfaces, each of which reveals a view of the band swaged into the band seat. The exposed surfaces shall be subjected to a light grinding operation to remove metal smear effects of saw cuts prior to measurements for depth of sidewall contact to determine compliance with the requirements.

4.5.4 Protective coating. The procedures for testing the cleaning of components and the application of protective coating shall be in accordance with the requirements of the applicable drawing and the specification listed thereon.

4.5.4.1 Protective coating - thickness. Protective coating thickness shall be accomplished using commercially avaliable equipment.

4.5.4.2 Salt spray testing. Salt spray testing shall be accomplished on production projectiles or sectioned projectiles that were acceptable at the time of painting and were sectioned subsequent to painting. If sectioned projectiles are used, sectioning shall be accomplished to allow testing of all major areas of the projectiles on an alternating basis. The salt spray test shall be performed utilizing the equipment and procedures specified by ASTM Method B-117 and TT-C-490 with the exception that test samples shall be selected in accordance with 4.4.3.5.1 and samples shall be dried in accordance with Drawing 12913927.

4.5.5 Radiographic inspection. Each shell for the first 1200 rounds shall be radiographed at least twice (0 and 90). A separate exposure may be required for adequate penetration of the base. If there are no failures in 1200 rounds then the radiographic inspection may be reduced to one random sample for every 25 thereafter. Any change in the process or a failure shall require an additional inspection of 1200 consecutive rounds. Penetrometers shall be used to verify 2% sensitivity on the bourrelet diameter and in the base area. The sensitivity of the radiographic technique shall be considered satisfactory when the image of the penetrometers, including the 2T holes, are perceptable in accordance with MIL-STD-00453. The penetrometers shall be made of steel having an approximate specific gravity of 7.8. For further details on construction of the penetrometers, see MIL-STD-00453.

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4.5.5.1 Check of photographic density of film. Film density shall be checked each hour of operation or fraction thereof, for each film processor. Greater frequency shall be required for any period during which difficulty is encountered in establishing the required densities. Photographic densities shall be measured with a diffuse density, photoelectric densitometer providing direct display of numerical density units. Accuracy of densitometer shall be maintained by calibrating with a photoelectric step tablet each hour immediately prior to measurements on the radiographic film. Each point of density measurement on the radiographs shall be identified by a circumscribed circle (made with a suitable pen or pencil) of no more than 1/4 inch in diameter. All photographic step tablets used in normal operation shall be checked for accuracy monthly by calibrating the densitometer with a reference photographic step tablet calibrated by National Bureau of Standards or approved by ARDEC. Referenced tablet shall be kept available, but shall be protected from wear and damage and shall not be used in daily operations. Any photographic step tablet having severe surface damage or showing a deviation from previous calibrations by .05D or more shall be replaced with a new, unused step tablet.

4.5.5.2 Review of radiographic sensitivity and techniques - (see 6.3). The sensitivity of the radiographic techniques shall be reviewed by submitting one radiograph of the projectile standard and a copy of radiographic procedures prior to the start of initial production. This radiograph shall be taken and processed using the same techniques as will be used in production.

4.5.5.3 Operator qualification. For operation of radiographic equipment, personnel shall be qualified and certified in accordance with MIL-STD-410, Level I. For set-up and calibration of radiographic equipment, personnel shall be qualified and certified in accordance with MIL-STD-410, Level II.

4.5.6 Hydrostatic test. The unpainted projectile body shall be mounted in a suitable test fixture and subjected to the required pressure. Outside surface of body shall not be restrained in any way during testing. Any body failing the requirement of 3.6.3a shall be rejected and removed from the lot.

4.5.6 Microstructure examination. A cross section shall be taken from each body subject to tensile testing for examination. The metallographic specimens shall be mounted, polished and etched per ASTM E-3 by suitable metallographic techniques so that an estimation of the size, distribution and form of graphic particles of each specimen can be made by a skilled metallographic or metallurgical engineer familiar with ductile iron. Characterization shall be per ASTM A-247.

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4.5.8 Proving ground functional tests.

4.5.8.1 Excess pressure. The ballistic test projectiles inert loaded and dummy fuzed will be fired in an acceptable M126 series cannon based upon current gun tube condemnation criteria. The propellant charge XM119E4 shall be adjusted to meet 54,000 psi plus or minus 2,000 psi. Photographic equipment (smear camera) will be set up near the gun muzzle to record the flight of the projectile. Visual observation will be made to assure projectile impacts in target area. Velocity and chamber pressures will be recorded.

5 PACKAGING

5.1 Packing and marking. The projectile metal parts shall be palletized in accordance with dwg. 12953254. The pallet shall be marked in accordance with dwg. 12953254. The packaging called out in this specification is for inert metal parts only.

6. NOTES.

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory).

6.1 Intended use. The projectile covered by this specification is intended for use as a target practice round for 155MM Howitzers.

6.2 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number and date of this specification.

b. Applicable stock number.

c. First article sample requirements.

d. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (See 2.1.2).

6.3 Submission of contractor inspection equipment designs for approval - See MIL-A-48078. Submit equipment designs as required to Commander, ARDEC, ATTN: SMCAR-QAR-I, Picatinny Arsenal, New Jersey 07806-5000. This address will be specified on the contract Data Requirements List, DD Form 1423 in the contract.

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6.4 Radiographic film. Only a film type and vendor as listed herein has been tested and approved by ARDEC for use in radiographing this projectile. A substitute item shall not be used before prior testing and approval by U.S. Army Armament Research, Development and Engineering Center, Picatinny Arsenal, New Jersey 07806-5000.

Eastman Kodak Company
1187 Ridge Road, West
Rochester, New York 14650

Film Types: AA, T, AX-1, or M

E.I. DuPont de Nemours and Company
Photo Products Department
Wilmington, Delaware 19898

Film Types: NDT-55, NDT-65 or NDT-70

GAF Corporation
140 West 51st Street
New York City, New York 10020

Film Types: 800 or 400

6.5 Inspection lot size. It is noted that the size of inspection lots of components, assemblies or items of delivery may differ from the actual quantities contractually scheduled for delivery. However, in order to facilitate scheduling of tests or deliveries, inspection lots of items of delivery may be equivalent to such contract quantities provided all of the lot formation criteria and sampling provisions of this specification are maintained.

6.6 Visual examination qualification. When compliance with the applicable requirement is in doubt as a result of visual examination, the characteristics may be measured or gaged to determine acceptability.

6.7 Submission of ballistic test data and ammunition data cards. In addition to the normal distribution of records for items procured by the Department of the Army, two copies of all ballistic test data and ammunition data cards shall be forwarded to Commander, ARDEC, ATTN: SMCAR-QAR-Q, and SMCAR-FSA-SS, Picatinny Arsenal, New Jersey 07806-5000 and AMSMC-QAD-D Army Master Depository, Rock Island, IL 61299.

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6.8 Alternative testing procedures. Alternative testing procedures may be used by the contractor when such testing and procedures provide as a minimum the quality assurance provisions specified herein. Prior to applying alternative procedures, the contractor shall describe them in a written proposal and shall demonstrate for the approval of the responsible Government technical agency that their effectiveness is equal to, or better than the quality assurance provisions specified herein.

6.9 Drawings. Drawings listed in Section 2 of the specification under the heading U.S. Army Armament Research, Development and Engineering Center (ARDEC), may also include drawings prepared by and identified as Edgewood Arsenal, Frankford Arsenal, Rock Island Arsenal, ARRADCOM, or Picatinny Arsenal drawings. Technical data originally prepared by these activities is now under the cognizance of ARDEC.

6.10 Process notes. The following notes are provided for information only:

a. The hydrostatic plug should be designed with a one quarter inch, minimum thread relief to prevent loading of initial threads on the warhead.

b. Examination should be made of all hydrostatic plugs after a warhead is blown during testing. Badly nicked or damaged plugs may cause destruction of acceptable items.

6.11 Submission of alternative quality conformance provisions. Unless otherwise specified in the contract, proposed alternative quality conformance provisions will be submitted by the contractor for evaluation by the technical activity responsible for the preparation of this specification.

6.12 Subject term (key word) listing.

Ammunition
Cast iron
Engineering testing
Inert metal parts
Target practice round

Custodian:
Army - AR

Preparing activity:
Army - AR

(Project 1320-AC60)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revisi letter should be given.
2. The submitter of this form must complete blocks 4, 5, 6, and 7.
3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization waive any portion of the referenced document(s) or to amend contractual requirements.

DOCUMENT IMPROVEMENT PROPOSAL	1. DOCUMENT NUMBER	2. DOCUMENT DATE (YYMMDD)
	MIL-P-70695 (AR)	15 NOVEMBER 1991

3. DOCUMENT TITLE
PROJECTILE, PRACTICE, 155MM, M804E1 CAST METAL PARTS ASSEMBLY

4. NATURE OF CHANGE *(Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets if needed.)*

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY

a. NAME	b. TELEPHONE (Include Area Code)	
US ARMY ARDEC STANDARDIZATION OFFICE	(1) Commercial (201) 724-6675	(2) AUTOVON 880-6675
c. ADDRESS (Include Zip Code) ATTN: SMCAR-BAC-S PICATINNY ARSENAL,NJ 07806-5000	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340	

DD Form 1426, OCT 89 *Previous editions are obsolete.* 198/2

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